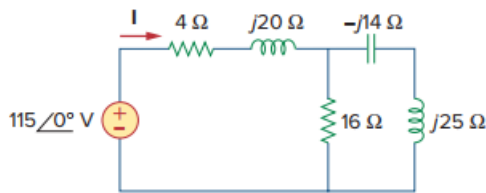


Problem

For the circuit shown in Fig. 9.46, find Z_{eq} and use that to find current I . Let $\omega = 10$ rad/s.

Figure 9.46:



Step-by-step solution

Step 1 of 2

Refer to Figure 9.46 in the text book for the circuit.

Determine the equivalent impedance Z_{eq} of the network.

$$\begin{aligned} Z_{eq} &= (4 + j20) + (16) \parallel (-j14 + j25) \\ &= (4 + j20) + \frac{(16) \times (j11)}{(16) + (j11)} \\ &= 9.135 + j27.47 \, \Omega \end{aligned}$$

Thus, the equivalent impedance, Z_{eq} of the circuit is $9.135 + j27.47 \, \Omega$.

[Comments \(4\)](#)



Anonymous

how do you simplify the second step?



Anonymous

No clue when this comment was posted but if others have the same question you would convert the numerator and denominator to polar coordinates, divide using polar rules, then convert back to rectangular.



Anonymous

how do you simplify the second step??



Anonymous

I figured it out without having to convert to Polar 1. First, multiply the top --> $(16 \times j11) = j176$ 2. Multiply the whole fraction by the complex conjugate -----> $(16 - j11)/(16 - j11)$ 3. You should get $(1986 + j2816)/(256 + 121)$ **Reminder $j^*j = -1$ and complex conjugate rule $(a + jb)(a + jb) = a^2 + b^2$. So $a = 16$ and $b = 11$ 4. You should have $(1936 + j2816)/(377)$ 5. Simplify --> $5.13 + j7.47$ 6. Add with $(4 + j20)$

Step 2 of 2

Determine the current from the source.

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}_{\text{eq}}}$$

Substitute $115\angle 0^\circ \text{ V}$ for $\mathbf{V}$ and $9.135 + j27.47 \Omega$ for $\mathbf{Z}_{\text{eq}}$.

$$\begin{aligned}\mathbf{I} &= \frac{115\angle 0^\circ}{9.135 + j27.47} \\ &= \frac{115\angle 0^\circ}{28.949\angle 71.6^\circ} \\ &= 3.972\angle -71.6^\circ \text{ A}\end{aligned}$$

Thus, the current $\mathbf{I}$ from the source is $\boxed{3.972\angle -71.60^\circ \text{ A}}$

Convert the current into time domain.

$$i(t) = 3.972 \cos(\omega t - 71.6^\circ) \text{ A}$$

Substitute 10 rad/s for ω .

$$i(t) = 3.972 \cos(10t - 71.6^\circ) \text{ A}$$

Thus, the current in time domain is $\boxed{3.972 \cos(10t - 71.6^\circ) \text{ A}}$.